

Assessing the Impact of Superspreaders on Epidemic Spread: A Reimplementation of the Fujie-Odagaki Model[★]

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Abstract

Severe Acute Respiratory Syndrome (SARS) demonstrated that epidemic outbreaks are strongly shaped by superspreaders, individuals who infect unusually many susceptible people. Because standard epidemiological models often assume uniform transmission rates across a population, they struggle to capture the sudden spikes caused by these events. This project reproduces the computational study by Ryo Fujie and Takashi Odagaki, which integrates superspreaders directly into a spatially structured Susceptible-Infected-Recovered (SIR) model [1]. Using Monte Carlo simulations in a two-dimensional continuous space, we evaluate two competing hypotheses: the “strong infectiousness” model and the “hub” model. Our reimplementation is consistent with the reference conclusion that the hub model, where superspreaders possess a wider radius of social connections, more closely explains the epidemic curves and secondary infection distributions observed during the 2003 SARS outbreak.

1. Introduction

The 2003 outbreak of Severe Acute Respiratory Syndrome (SARS) was one of the clearest examples of how a respiratory disease can spread unevenly through a population. SARS was first identified in Guangdong, China, in November 2002, and during the following months it spread across Asia, Europe, and the Americas. By the end of the epidemic, more than 8,000 people worldwide had been infected and more than 700 people had died [2]. The outbreak therefore provides an important case study for mathematical modelling because its spread was rapid, spatially uneven, and strongly affected by individual-level differences in transmission.

A central feature of the SARS outbreak was the role of “superspreaders.” According to the United States Centers for Disease Control and Prevention (CDC), a superspreader is a patient who directly infects more than ten other people [3]. Reports from Hong Kong and Singapore showed that these superspreading events were responsible for sharp increases in patient numbers [3, 5, 6]. In other words, the epidemic did not grow only through a smooth average transmission process; instead, a small number of infected individuals caused unusually large clusters of secondary cases.

This behavior creates a problem for traditional epidemic models. Standard mathematical models such as the SIR (Susceptible - Infected - Recovered) model divide the population into uniform compartments and usually assume that transmission rates between all pairs of individuals are equal [4]. While this assumption makes the model mathematically simple, it removes the biological and social heterogeneity that is essential for understanding superspreading. In particular, a uniform SIR model cannot easily explain why the SARS outbreaks in Hong Kong and Singapore included sudden jumps in case numbers caused by a small subset of patients.

[★]Source Code Repository: <https://github.com/tuananhlevan/math-modelling-project>

To address this limitation, the standard SIR model must be generalized to incorporate the direct effects of superspreaders. In their foundational paper, *Effects of superspreaders in spread of epidemic*, Fujie and Odagaki proposed an alternative approach: placing individuals in a continuous two-dimensional space where the probability of infection depends heavily on spatial distance, allowing for distinct transmission rules for normal individuals versus superspreaders [1]. This report details our computational reimplementations of this spatial model, exploring whether a superspreader’s impact is driven by intrinsically stronger infectiousness or by a broader network of social hubs.

2. Modelling Approach

Fujie and Odagaki chose to generalize the standard SIR model by embedding it in a two-dimensional continuous spatial structure. Instead of describing the epidemic only through deterministic differential equations for the total numbers of susceptible, infected, and recovered individuals, the model treats people as individual agents located at fixed coordinates. The epidemic then evolves through stochastic Monte Carlo infection events between infected and susceptible individuals.

The motivation for this choice becomes clearer when compared with the classical SIR model. In a closed population, the standard deterministic formulation assumes

$$N = S + I + R, \quad (1)$$

where S , I , and R are the numbers of susceptible, infected, and recovered individuals. Its dynamics are usually written as

$$\frac{dS}{dt} = -\beta \frac{SI}{N}, \quad (2)$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I, \quad (3)$$

$$\frac{dR}{dt} = \gamma I. \quad (4)$$

Here β is the transmission rate and γ is the recovery rate. These equations are useful because they are compact and analytically interpretable, but they also impose strong idealizations. The term SI/N assumes a well-mixed population, meaning that every susceptible individual can effectively encounter every infected individual. Geographical distance is ignored, and all infected individuals are treated as equally infectious through the same constant transmission rate β .

These assumptions are too restrictive for SARS superspreading. Superspreading events suggest that transmission depends on both spatial contact structure and individual heterogeneity. Fujie and Odagaki address the first limitation by placing individuals in a continuous two-dimensional domain and making infection probability a function of distance r . They address the second limitation by assigning a subset of individuals as superspreaders with different transmission rules from normal individuals.

The paper therefore proposes two competing superspreader interpretations. The *strong infectiousness model* treats superspreaders as intrinsically more infectious at the same contact range. The *hub model* treats superspreaders as individuals with many more social contacts or a wider effective contact radius. Comparing these two models allows the study to ask whether SARS superspreading is better explained by biological infectiousness or by social connectivity.

Our reimplementations follow this modelling logic in code. In `src/individual.py`, each person is represented by an `Individual` object with an explicit coordinate pair `[x, y]`, a health state, and a `super_spreader` boolean. The coordinate pair implements the spatial component of the model, while the superspreader flag allows the infection rule to change depending on the type of infected individual. The `infect` method computes the distance between two individuals and applies the appropriate probability function for the selected model, directly mirroring the agent-based Monte Carlo structure of the reference paper.

3. Mathematical Model

The model consists of N individuals distributed in an $L \times L$ continuous two-dimensional space. The individuals have fixed positions during a simulation. The reference model imposes periodic boundary conditions to reduce artificial edge effects; in this reimplementation, horizontal wrapping is used while the vertical direction is kept open so that bottom-to-top percolation can be measured. One initial infected individual is placed at the bottom of the system, while the remaining $N - 1$ individuals are distributed randomly and begin in the susceptible state. A fraction λ of the population is assigned as superspreaders, while the rest are normal individuals.

Each individual is in one of three epidemiological states: susceptible (S), infected (I), or recovered (R). The state transition follows

$$S \longrightarrow I \longrightarrow R, \quad (5)$$

but the infection step is stochastic and spatial. If an infected individual and a susceptible individual are separated by distance r , then the susceptible individual becomes infected with probability $w(r)$. After attempting transmission, the infected individual recovers with probability γ . In the reference simulations, Fujie and Odagaki use $L = 10r_0$, $w_0 = 1$, $\gamma = 1$, and population sizes from $N = 150$ to 900 , with results averaged over 1000 Monte Carlo runs.

For normal infected individuals, the infection probability decreases quadratically with distance and becomes zero outside the cutoff radius r_0 :

$$w_{\text{normal}}(r) = \begin{cases} w_0 \left(1 - \frac{r}{r_0}\right)^2, & 0 \leq r \leq r_0, \\ 0, & r > r_0. \end{cases} \quad (6)$$

Equivalently, inside the cutoff the probability is proportional to $(r - r_0)^2$, scaled by w_0/r_0^2 . This assumption makes normal transmission local: nearby susceptible individuals face a higher risk, while individuals beyond r_0 cannot be infected by that contact.

The first superspreader hypothesis is the *strong infectiousness model*. In this model, a superspreader is intrinsically more infectious, but its transmission radius remains the same as that of a normal individual:

$$w_{\text{strong}}(r) = \begin{cases} w_0, & 0 \leq r \leq r_0, \\ 0, & r > r_0. \end{cases} \quad (7)$$

Thus $w(r)/w_0 = 1$ throughout the entire radius r_0 . Epidemiologically, this model represents a person with unusually high biological infectiousness, such as high viral load, or a confined environment where distance loses much of its protective effect. Any susceptible person inside the radius faces the maximum infection probability.

The second superspreader hypothesis is the *hub model*. In this model, superspreaders do not transmit with a constant probability. Instead, their infection probability decays with distance exactly like a normal individual's, but their effective reach is extended to a larger cutoff radius r_n :

$$w_{\text{hub}}(r) = \begin{cases} w_0 \left(1 - \frac{r}{r_n}\right)^2, & 0 \leq r \leq r_n, \\ 0, & r > r_n. \end{cases} \quad (8)$$

For normal individuals $r_n = r_0$, while for superspreaders in the hub model $r_n = \sqrt{6}r_0$. This model represents social rather than biological heterogeneity: a superspreader may be a healthcare worker, taxi driver, public transport operator, or community hub who interacts with many more susceptible individuals over a wider area.

The factor $\sqrt{6}$ is chosen so that the total expected infection capacity of a hub superspreader equals that of a strong-infectiousness superspreader. In a spatially uniform population with density ρ , the expected number of secondary infections in one time step is proportional to the integral of $w(r)$ over the two-dimensional contact region:

$$I = \int_0^{2\pi} \int_0^{R_{\text{max}}} w(r) r dr d\theta = 2\pi \int_0^{R_{\text{max}}} w(r) r dr. \quad (9)$$

For the strong infectiousness model,

$$I_{\text{strong}} = 2\pi \int_0^{r_0} w_0 r dr = \pi w_0 r_0^2. \quad (10)$$

For the hub model,

$$I_{\text{hub}} = 2\pi \int_0^{r_n} w_0 \left(1 - \frac{r}{r_n}\right)^2 r dr \quad (11)$$

$$= 2\pi w_0 \left[\frac{r^2}{2} - \frac{2r^3}{3r_n} + \frac{r^4}{4r_n^2} \right]_0^{r_n} \quad (12)$$

$$= \frac{\pi w_0 r_n^2}{6}. \quad (13)$$

Equating I_{strong} and I_{hub} gives

$$\pi w_0 r_0^2 = \frac{\pi w_0 r_n^2}{6} \Rightarrow r_n = \sqrt{6} r_0. \quad (14)$$

This derivation shows that the two superspreader models are normalized to have the same total potential outbreak power. Therefore, differences in simulation results are caused mainly by the spatial distribution of contacts, not by giving one model a larger total infectivity budget.

Our code maps these mathematical rules directly in the `Individual.infect()` method in `src/individual.py`. When `model_type == 1`, the code applies the strong infectiousness rule: if the infected individual is a superspreader and $r \leq r_0$, the infection probability is set to the constant value `SUPER_INFECT`; otherwise, normal individuals use the distance-decay formula. When `model_type == 2`, the code applies the hub rule: superspreaders use the same quadratic distance-decay formula, but the cutoff radius is expanded to `INFECT_CUTOFF * sqrt(6)`. Thus, the implementation mirrors the two hypotheses in the reference paper.

Several output quantities are measured from these dynamics. The *percolation probability* is the fraction of trials in which an infection starting near the bottom reaches the top of the system. The *critical density* is the population density at which this transition from local extinction to system-wide spread occurs. The *epidemic curve* records the number of infected individuals over time, while the distribution of secondary infections records how many people are directly infected by each infected case.

4. Solving Algorithm

The model is solved by Monte Carlo simulation. Each simulation advances in discrete time steps. At each step, the currently infected individuals attempt to infect susceptible individuals using the distance-dependent probability $w(r)$. Successfully infected individuals transition to the infected state, but they do not attempt transmission until the next time step. The original infected individuals recover with probability γ , which is set to 1.0 in this study. The simulation also tracks whether the infection percolates, meaning that an outbreak starting at the bottom of the spatial domain reaches the top.

Algorithm 1 Monte Carlo simulation for one parameter setting

Require: Superspreader probability λ , population size N , model type M

```
1: Generate  $N$  individuals in the square domain
2: Place patient zero at the bottom center and set its state to infected
3: Mark each individual as a superspreader with probability  $\lambda$ 
4:  $percolated \leftarrow false$ 
5: while at least one individual is infected do
6:    $I_t \leftarrow$  all individuals infected at the current time step
7:    $S_t \leftarrow$  all susceptible individuals
8:   for each infected individual  $i \in I_t$  do
9:     for each susceptible individual  $s \in S_t$  do
10:      Compute the distance  $r$  between  $i$  and  $s$ 
11:      Compute  $w(r)$  according to model type  $M$ 
12:      if a random number in  $[0, 1]$  is less than  $w(r)$  then
13:        Set  $s$  to infected
14:        if  $s$  reaches the top percolation threshold then
15:           $percolated \leftarrow true$ 
16:        end if
17:      end if
18:    end for
19:    Recover  $i$  with probability  $\gamma$ 
20:  end for
21: end while
```

Repeating this procedure many times is essential because the model is stochastic. A single trial may die out by chance before reaching the top boundary, or it may percolate by creating a connected chain of infections from the bottom to the top of the population. In the implementation, percolation is recorded as a boolean outcome, but the time-step loop continues until there are no infected individuals left. Averaging over many runs gives stable estimates of percolation probability, propagation speed, epidemic shape, and the distribution of secondary cases.

In our reimplementaion, this logic is implemented in `src/engine.py`. The `main_loop()` function builds lists of currently infected and currently susceptible individuals at every time step. It then calls `infect()` for each infected - susceptible pair and `recover()` for each processed infected individual. Newly infected individuals are counted for the epidemic curve. The function also checks whether any newly infected individual's vertical coordinate is above the percolation threshold; if so, it sets the percolation boolean to `True`. This directly implements the paper's percolation definition while still allowing the simulated outbreak to finish naturally.

5. Reference Results and Interpretation

The reference study evaluates the models through several complementary quantities rather than by relying on a single final comparison. In our reimplementaion, we regenerated the same categories of plots: percolation probability, critical density, propagation speed, epidemic curves, secondary infection distributions, and comparison with the Singapore SARS data. These quantities can be defined mathematically from the Monte Carlo outputs, which makes the comparison more precise.

5.1. Output Metrics

Let K be the number of Monte Carlo runs and let X_k be the percolation indicator for run k :

$$X_k = \begin{cases} 1, & \text{if the infection reaches the top boundary,} \\ 0, & \text{otherwise.} \end{cases} \quad (15)$$

The estimated percolation probability is therefore

$$\hat{P}_{\text{perc}} = \frac{1}{K} \sum_{k=1}^K X_k. \quad (16)$$

The critical density ρ_c is the density at which this probability crosses a chosen threshold, commonly interpreted as the transition point between extinction and system-wide spread:

$$\rho_c = \inf\{\rho : \hat{P}_{\text{perc}}(\rho) \geq p^*\}, \quad (17)$$

where p^* is a fixed reference probability such as 0.5. Fujie and Odagaki also interpret this transition using a basic reproductive number,

$$R_0(\lambda) = \frac{\rho}{w_0} \left[\lambda \int w_{\text{ss}}(r) 2\pi r dr + (1 - \lambda) \int w_n(r) 2\pi r dr \right], \quad (18)$$

where w_{ss} and w_n are the infection-probability functions for superspreaders and normal individuals. The critical-density curves in Figure 2 compare the simulated transition with the reference condition $R_0 = R_c$, where $R_c = 4.5$ for the strong infectiousness model and $R_c = 3.2$ for the hub model [1, 8].

The epidemic curve records the average number of newly infected individuals at each time step. If $C_k(t)$ is the number of new infections at time t in run k , then

$$\bar{C}(t) = \frac{1}{K} \sum_{k=1}^K C_k(t). \quad (19)$$

Similarly, if $R_k(t)$ is the front-line distance reached by the infection from patient zero in run k , the average propagation distance is

$$\bar{R}(t) = \frac{1}{K} \sum_{k=1}^K R_k(t), \quad (20)$$

where the front line is measured before the boundary-induced plateau. In the reimplementaion, this distance uses horizontal wrapping and an open vertical direction, matching the bottom-to-top percolation geometry. The propagation speed can then be approximated by a finite difference,

$$v(t) \approx \frac{\bar{R}(t+1) - \bar{R}(t)}{\Delta t}. \quad (21)$$

These definitions clarify what each plot measures: percolation probability captures final spatial reach, the epidemic curve captures temporal outbreak intensity, and the secondary infection distribution captures individual-level transmission heterogeneity.

5.2. Percolation and Critical Density

The first important result is the percolation transition. Figure 1 shows regenerated percolation probability curves for the strong infectiousness and hub models. In both cases, the probability is close to zero at low population density and rises toward one once the density becomes sufficiently large. This reproduces the qualitative result of Fujie and Odagaki: epidemic spread behaves like a threshold phenomenon. Below the threshold, infected individuals usually

recover before the infection chain can reach the top of the system. Above the threshold, there are enough reachable susceptible individuals for the infection path to span the population.

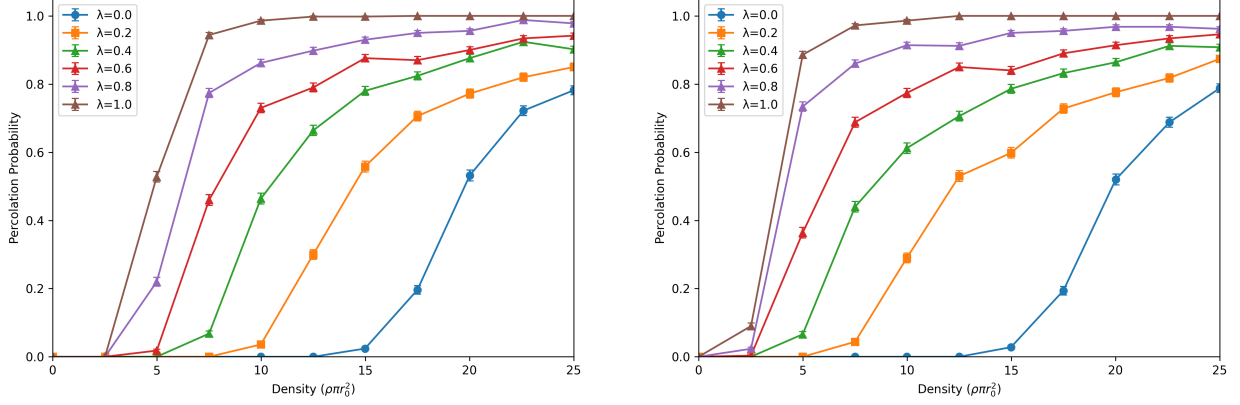


Figure 1: Reimplemented percolation probability curves for the strong infectiousness model (left) and hub model (right)

The critical-density plot in Figure 2 summarizes this threshold behavior more directly. As the superspreader fraction λ increases, the critical density decreases. This means that superspreaders make large outbreaks possible even when the population is not very dense. The hub model generally has a lower critical density than the strong infectiousness model because long-range contact radius is more effective at connecting spatially separated clusters. Thus, even when the two models are normalized to have comparable expected infection strength, the spatial placement of contacts still matters.

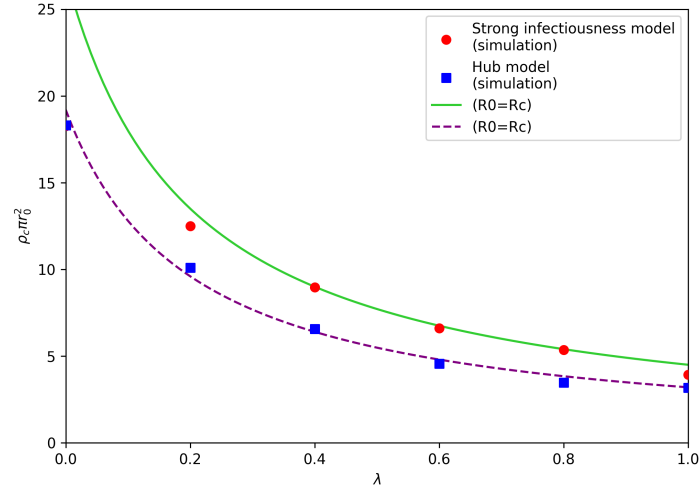


Figure 2: Reimplemented critical density as a function of superspreader fraction

5.3. Spatial Routes of Infection

To intuitively grasp the difference between the superspreader mechanisms, we can visualize the explicit routes of infection across the spatial domain. Figures 3, 4, and 5 illustrate a sample outbreak for each model. In the normal model, the infection spreads through short, localized steps. In contrast, the strong infectiousness model generates dense, localized clusters of infections around superspreaders. The hub model, however, produces long-range infection

links that bridge geographically distant clusters, visibly explaining why it achieves a lower critical density for system-wide percolation.

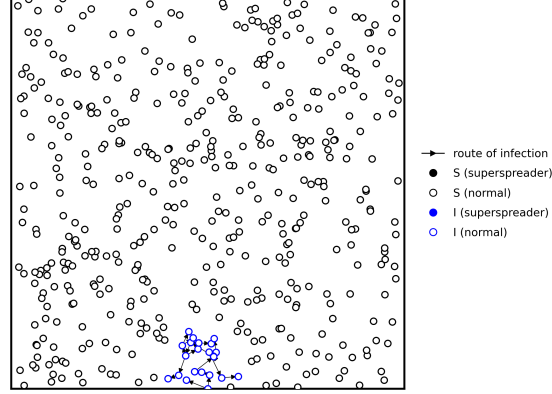


Figure 3: Reimplemented route of infection for the normal model ($\lambda = 0.0$)

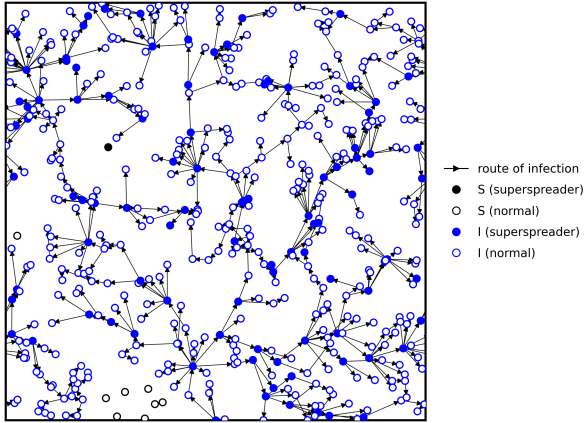


Figure 4: Strong infectiousness ($\lambda = 0.2$)

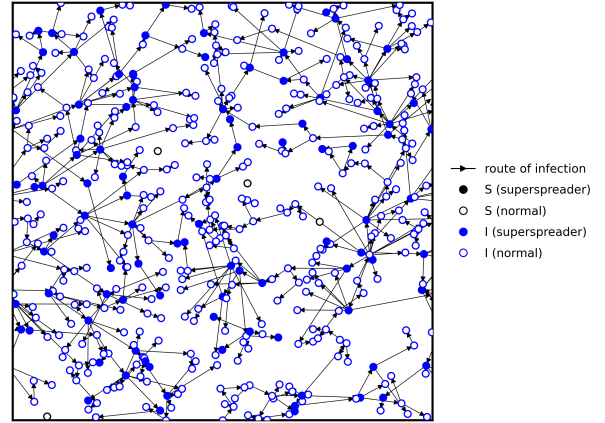


Figure 5: Hub model ($\lambda = 0.2$)

5.4. Propagation Speed and Epidemic Curves

The propagation and epidemic-curve plots provide a temporal interpretation of the same mechanism. Figure 6 shows the propagation distance over time, where higher superspreader fractions lead to much faster spatial advancement. Similarly, in Figure 7, the no-superspreader curve is comparatively slower and smoother, while the curves with superspreaders rise sharply and reach a higher peak.

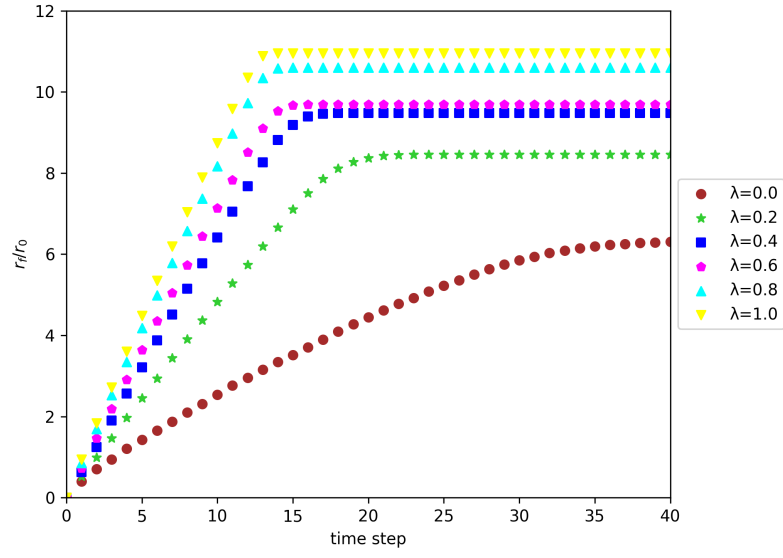


Figure 6: Reimplemented propagation distance over time for varying superspreader fractions

This matches the reference observation that superspreaders create sudden increases in case numbers. The peak is not simply larger; it is also more concentrated in time, which is important for SARS because the observed outbreak curves contained abrupt surges rather than slow uniform growth.

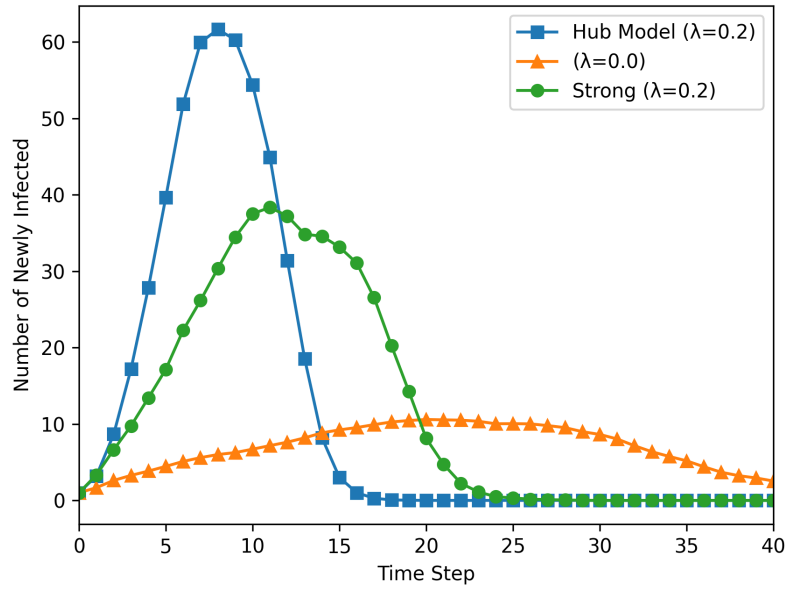


Figure 7: Reimplemented epidemic curves at $\rho\pi r_0^2 = 20$

5.5. Secondary Infections and SARS Comparison

Another important result is the distribution of secondary infections. Figure 8 shows that when superspreaders are included, the distribution develops a long right tail. Many infected individuals cause few or no further infections, but a small number infect many people. This is exactly the type of heterogeneity that a classical well-mixed deterministic SIR model cannot describe at the individual level. The long-tailed distribution supports the modelling choice: superspreading should be represented as individual-level variation in transmission opportunities rather than only as a change in the average transmission rate.

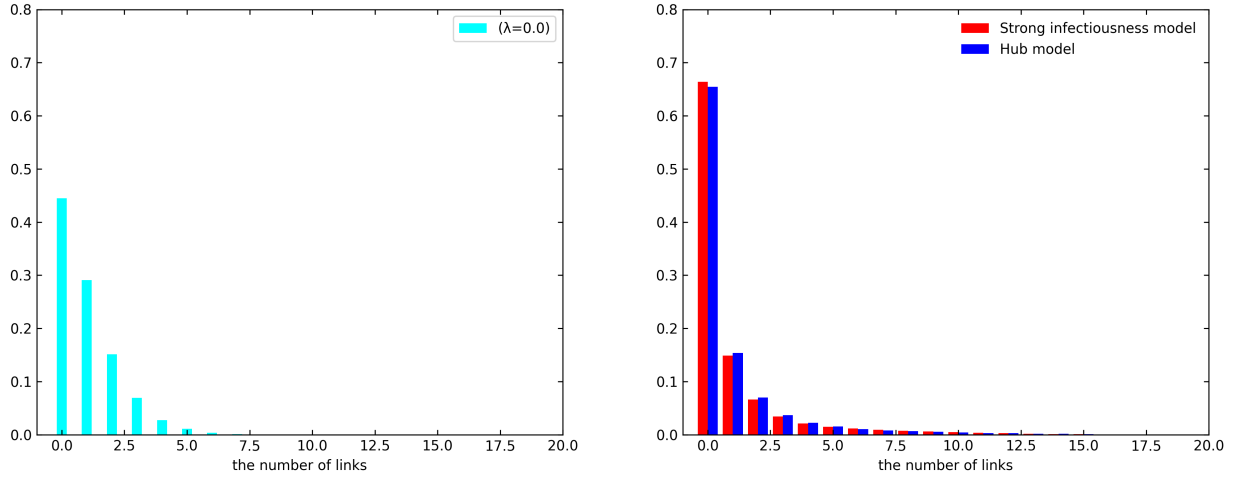


Figure 8: Reimplemented secondary infection distributions at $\rho\pi r_0^2 = 15$

To compare this with historical data, Figure 9 shows the empirical distribution of secondary infections from the 2003 Singapore SARS outbreak. The empirical distribution exhibits the exact same long right tail observed in our superspreader simulations, validating the necessity of modeling individual-level heterogeneity.

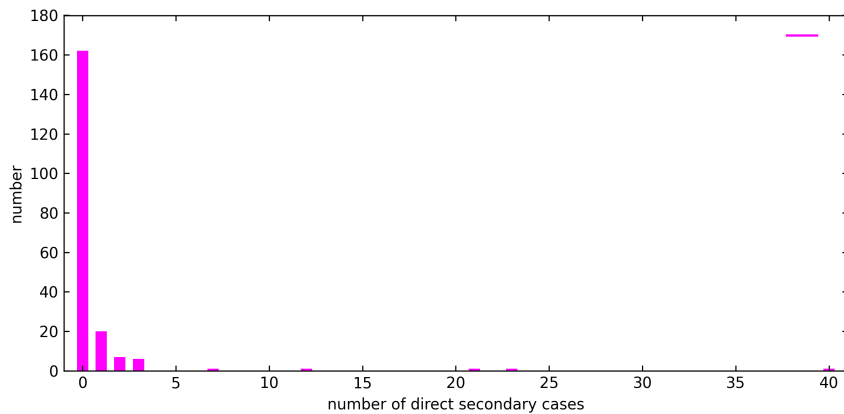


Figure 9: Empirical distribution of secondary cases from the Singapore SARS outbreak

Finally, Figure 10 compares the simulated epidemic curves with the Singapore SARS epidemic curve used in the reference paper. The empirical secondary infection data include individuals who directly infected 12, 21, 23, and 40

people (as shown in Figure 9), and the epidemic curve has a rapid rise followed by decline [3, 9]. Our regenerated comparison is consistent with the same conclusion as Fujie and Odagaki: although both superspreader mechanisms improve over the no-superspreader model, the hub model better captures the shape of the observed SARS curve. This suggests that the most important feature of SARS superspreaders was not only stronger local infectiousness, but a wider effective contact network.

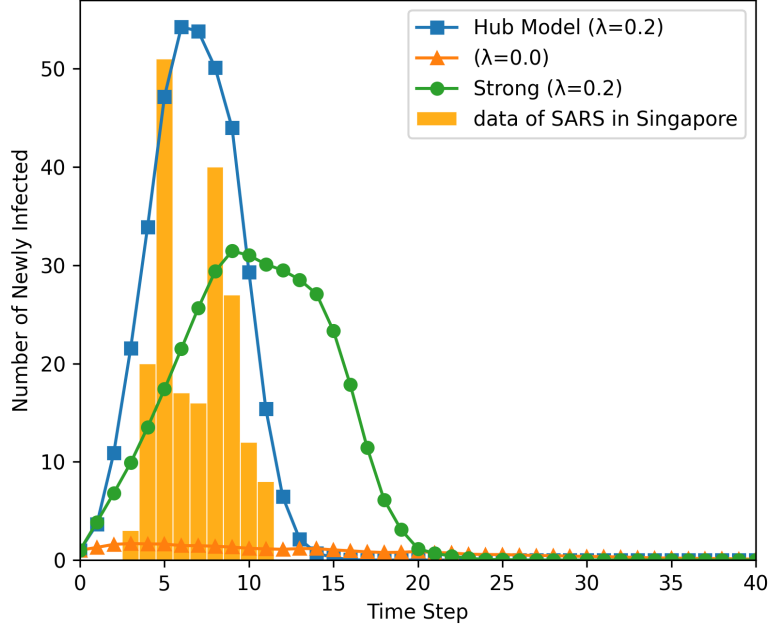


Figure 10: Reimplemented comparison between the simulated epidemic curves and the Singapore SARS epidemic curve

Taken together, these reimplemented results support the reference paper across more than the final SARS comparison alone. The percolation plots show that superspreaders lower the outbreak threshold, the epidemic curves show that they create sharper temporal peaks, and the secondary infection distributions show that they generate the observed heterogeneity in individual transmission. The final comparison with SARS then identifies which superspreader mechanism is more plausible. The hub model is favored because long-range or high-degree social connectivity explains both faster spatial propagation and the sharp epidemic curve, while the strong infectiousness model mainly increases local transmission intensity.

6. Sensitivity Analysis and Robustness

In this section, we evaluate the structural robustness of the spatial model and its sensitivity to the introduction of superspreaders. True robustness is demonstrated when a model’s macroscopic behavior remains consistent with its governing physical theories, even when significant microscopic perturbations are introduced. We then separate this structural robustness from the model’s sensitivity, which measures how drastically the epidemic outcomes react to changes in individual transmission heterogeneity.

6.1. Robustness of the Macroscopic Transition

The primary demonstration of the model’s robustness is found in Figure 2. The figure plots the simulated critical density ρ_c across different superspreader fractions λ . The theoretical continuum curves $R_0 = R_c$ represent the basic reproductive number required for a sustained outbreak in a continuous spatial domain.

As λ shifts from 0 to 1, we fundamentally alter the agent-level transmission mechanics by distributing a large portion of the infection capacity to a small subset of the population. Despite this extreme microscopic heterogeneity, the critical density extracted from the Monte Carlo simulation strictly adheres to the theoretical continuum curve $R_0 = R_c$ for both the strong infectiousness and hub models.

This adherence proves that the observed percolation threshold is not an artifact of the agent-based simulation code, but a robust feature of the underlying spatial epidemic theory. The model’s core spatial structure is mathematically robust against extreme changes in population heterogeneity, maintaining a predictable phase transition governed by the total effective R_0 .

Furthermore, this robustness is supported by the normalization derivation $r_n = \sqrt{6}r_0$. The hub model is not simply given an unfairly larger total infection strength; its total expected capacity is normalized to match the strong infectiousness model. Thus, the conclusion that the hub geometry (wider contact reach) more closely matches the SARS data is a robust structural finding, not a consequence of parameter tuning.

6.2. Sensitivity to Superspreaders

While the model is structurally robust, its epidemic outcomes are highly sensitive to the presence of superspreaders. A simple way to express this sensitivity is by comparing two nearby values of an input parameter. If θ is the changed variable and Y is the measured output, the local finite-difference sensitivity is

$$S_{\theta,Y}(\theta_j) \approx \frac{Y(\theta_{j+1}) - Y(\theta_j)}{\theta_{j+1} - \theta_j}. \quad (22)$$

Using this formulation on the data in Figure 2, we can measure how the critical density responds to the introduction of superspreaders. When λ increases from 0.0 to 0.2, the sensitivity of the critical density is approximately -41.10 for the hub model and -28.87 for the strong infectiousness model. This steep initial drop demonstrates that the threshold for large outbreaks is highly sensitive to the initial presence of superspreaders. The model is also sensitive at the individual level: as shown in the secondary infection distributions, introducing superspreaders creates a long right tail of transmission events, fundamentally shifting the epidemic dynamics away from the uniform average.

Finally, the statistical sensitivity to the Monte Carlo simulation’s intrinsic randomness is minimal. The percolation probability plotted in Figures 1 is a stochastic estimator

$$\hat{P}_{\text{perc}} = \frac{1}{K} \sum_{k=1}^K X_k, \quad \text{with uncertainty } SE(\hat{P}_{\text{perc}}) = \sqrt{\frac{\hat{P}_{\text{perc}}(1 - \hat{P}_{\text{perc}})}{K}}. \quad (23)$$

This uncertainty is visually incorporated into the percolation plots as standard error bars. With $K = 1000$ runs per configuration, the maximum standard error is $\sqrt{0.5 \times 0.5 / 1000} \approx 0.0158$. The negligible size of these error bars demonstrates that the sharp phase transition is statistically stable against stochastic noise.

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